COLUMBIA UNIVERSITY
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STAT 5703

Statistical Inference and Modeling for Data Science

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Regression Model Assumptions

Recall the multiple linear regression model:

$$\boldsymbol{Y} = \boldsymbol{X\beta} + \boldsymbol{\varepsilon}$$

The assumption is that $\boldsymbol{\varepsilon} \sim N_n(0, \sigma^2 \boldsymbol{I})$, which means

$$\boldsymbol{Y}|\boldsymbol{X} \sim N_n(\boldsymbol{X\beta}, \sigma^2 \boldsymbol{I})$$

The above implies several things about the errors:

- Normality
- Independence (because uncorrelated and normal)
- Same variance
- The model also implies linearity, in the sense $\mathrm{E}(\boldsymbol{Y}|\boldsymbol{X}) = \boldsymbol{X\beta}$

- How do we check them and what if they are violated?

Non-Normal Errors Detection

- Box-Plot of Residuals – Can be used to check for symmetry and for outliers
- Check proportion that lie within 2 standard deviations around 0, ±2*SD, where SD=sqrt(MSE)
- Normal probability plot of residual versus expected values under normality – should fall approximately on a straight line (Only works well with moderate to large samples) **qqnorm(e); qqline(e)** in R

Expected value of Residuals under Normality:

1) Rank residuals from smallest (large/negative) to highest (large/positive) Rank = k

2) Compute the percentile using $p = \dfrac{k - 0.375}{n + 0.25}$ and obtain corresponding z-value: $z(p)$

3) Multiply by $s = \sqrt{MSE}$ expected residual = $\sqrt{MSE}\left[z(p)\right]$

Tests for Normality of Residuals

- Shapiro-Wilk Test –involves comparing the observed data with a corresponding set of normally distributed values. The *W* statistic is derived from the correlation between the data and the corresponding normal scores. A value of *W* close to 1 suggests that the data follow a normal distribution. The test is known for its good power properties, meaning it has a high probability of correctly detecting departures from normality even in small samples. In R: use shapiro.test
- Kolmogorov – Smirnov test – a general test for comparing a sample to any distribution. In R: use ks.test

What if we drop the normality assumption?

There are several alternatives:

- Assume another distribution (GLM models)
- Transform the data to look "normal"
- Fit a nonparametric regression.
- One example of nonparametric model is the so-called Theil's method.
- It still assumes the linearity condition, but normality is replaced with
- The conditional distribution of $Y|x$ is symmetric and continuous.
- The estimation is done by ranking, like the Wilcoxon test.

Model Departures Detected With Residuals and Plots

- Relation between Y and X is not linear
- Errors have non-constant variance
- Errors are not independent
- Existence of Outlying Observations
- Non-normal Errors
- Missing predictor variable(s)
- Common Plots
  - Residuals/Absolute Residuals versus Predictor Variable
  - Residuals/Absolute Residuals versus Predicted Values
  - Residuals versus Omitted variables
  - Residuals versus Time
  - Box Plots, Histograms, Normal Probability Plots

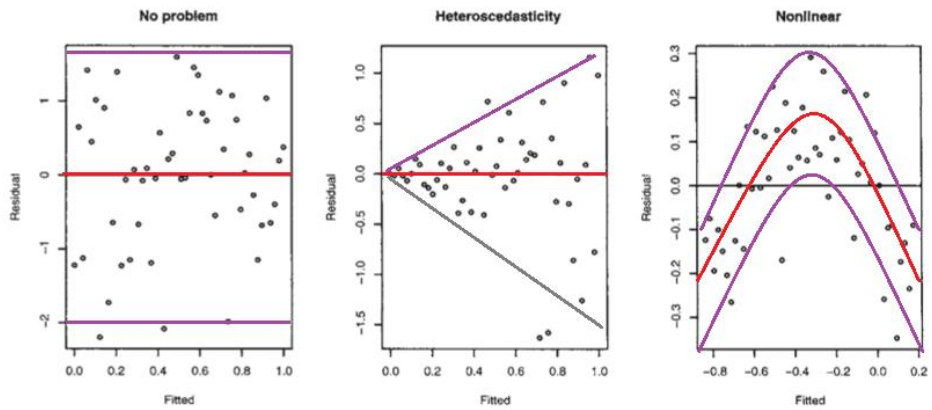
Diagnostics Plots

- The most common diagnostic plot is the *residual plot*:

Plot (standardized) residuals versus *X* or residuals vs. predicted values

What to look for?

- Random Cloud around 0 $\Rightarrow$ all good (first panel)
- Funnel Shape $\Rightarrow$ Non-constant Variance (second panel)
- Curved pattern (u-shape) $\Rightarrow$ Nonlinearity (third panel)

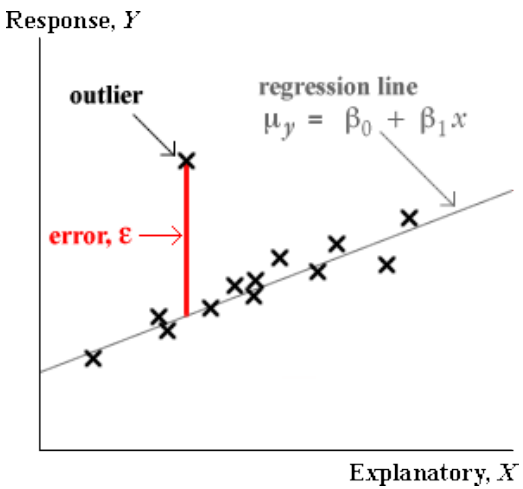


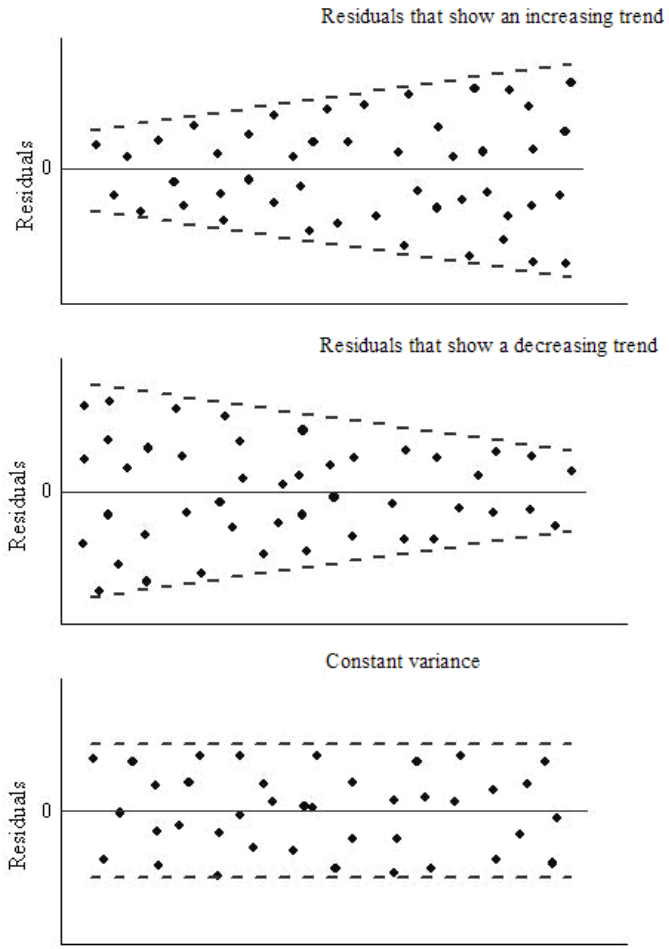
You can also check for outliers: fall far above (positive) or below (negative) the general cloud pattern

Diagnostics Plots

- Residual plots:

A closer look at the non-constant variance:

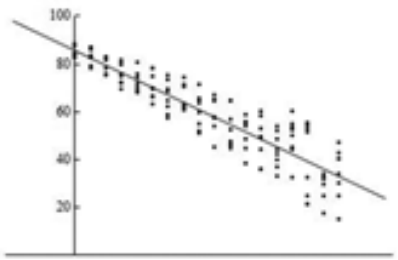




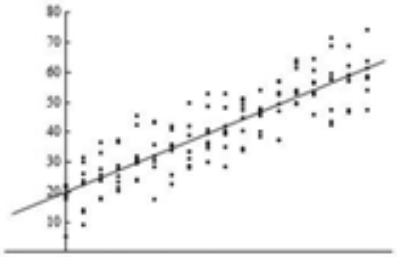
Diagnostics Plots: Try it

Below are four scatterplots with regression lines shown and four corresponding residual plots. Match the scatterplot to its residual plot.

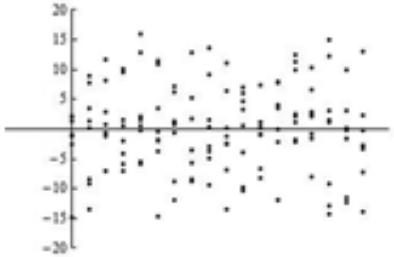
Scatterplot I



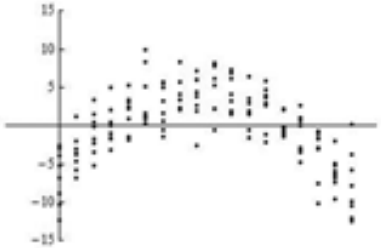
Scatterplot II



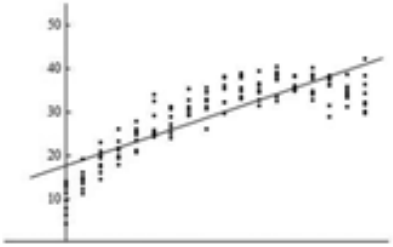
Residual Plot A



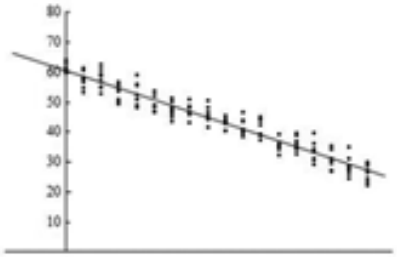
Residual Plot B



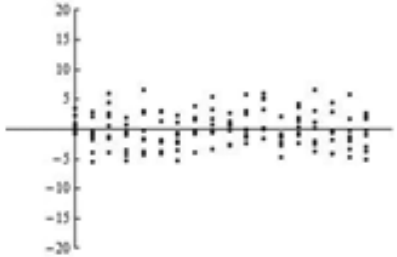
Scatterplot III



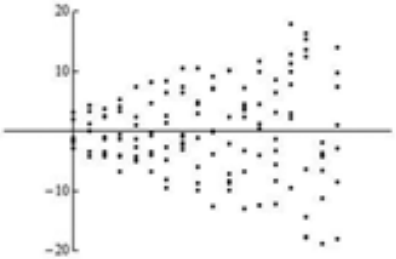
Scatterplot IV



Residual Plot C



Residual Plot D



Linear vs. Nonlinear Models

Suppose we have a simple regression with

$$E(Y|X = x) = f(x, \boldsymbol{\beta})$$

- Classic linear model: the function $f(x, \boldsymbol{\beta})$ is linear in *both* x and $\boldsymbol{\beta}$. For example,

$$f(x, \beta) = \beta_0 + \beta_1 x$$

We know this because the partial derivatives $\frac{\partial f(x,\boldsymbol{\beta})}{\partial x}, \frac{\partial f(x,\boldsymbol{\beta})}{\partial \beta_0}, \frac{\partial f(x,\boldsymbol{\beta})}{\partial \beta_1}$ do not depend on the variable we differentiate with respect to. For example, $\frac{\partial f(x,\boldsymbol{\beta})}{\partial x} = \beta_1$ is free of the x variable. Can be estimated with simple LS method straightaway.

- Nonlinear with respect to x *only*. For example, the quadratic polynomial model:

$$f(x, \boldsymbol{\beta}) = \beta_0 + \beta_1 x + \beta_2 x^2$$

Then $\frac{\partial f(x,\boldsymbol{\beta})}{\partial x} = \beta_1 + \beta_2 x$ *depends* on x. Can be estimated with multiple LS after renaming the quadratic predictor to $x_2 = x^2$.

- Nonlinear with respect to the *parameters* (and x). For example:

$$f(x, \boldsymbol{\beta}) = \beta_0 + \beta_1 e^{-\beta_2 x}$$

Then $\frac{\partial f(x,\boldsymbol{\beta})}{\partial x}$ depends on x and $\frac{\partial f(x,\boldsymbol{\beta})}{\partial \beta_2}$ depends on β_2. Can <u>not</u> be estimated with LS!

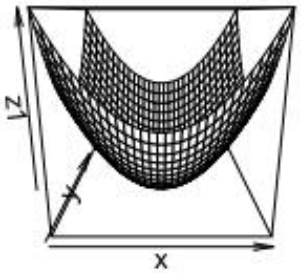
Polynomial Regression Models

- What is polynomial regression?
  - A form of regression analysis in which the relationship between the independent variables and the dependent variable is modeled as a polynomial of degree *p*.
  - Fits a nonlinear relationship between *x* and E(*Y* | *x*).
  - Usually fit with the method of least squares.
- Models with one predictor: Using a polynomial of degree *p* creates *p* - 1 "bends". For example:
  - 2nd order model: $E(Y \mid X) = \beta_0 + \beta_1 X + \beta_2 X^2$ has one bend
  - 3rd order Model: $E(Y \mid X) = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3$ has two bends
- Response surface models with two (or more) predictors
  - 2nd order model with two predictors:

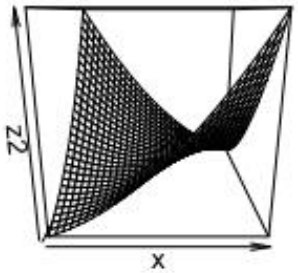
$$E(Y|X_1, X_2) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1^2 + \beta_4 X_2^2 + \beta_5 X_1 X_2$$

Response Surface Examples

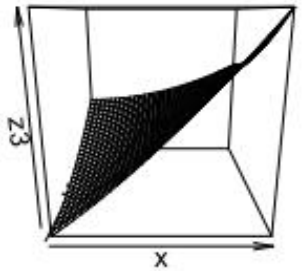
$x^2 + y^2$



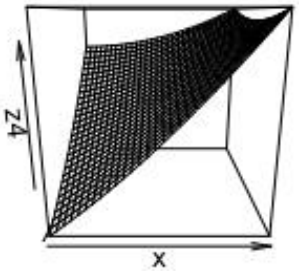
$x^2 + y^2 - 3xy$



$5x + 5x^2 + y^2 - 3xy$



$5x + 3y + x^2 + y^2 - 3xy$



Test For Independence - Durbin-Watson Test

Consider the following linear regression model:

$$Y_i|x_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$
$$\varepsilon_i = \rho\varepsilon_{i-1} + u_i$$
$$u_i \sim N(0, \sigma^2), |\rho| < 1$$

We want to test the hypotheses:

$$H_0: \rho = 0 \Rightarrow \text{errors are uncorreleated}$$
$$H_0: \rho \neq 0 \Rightarrow \text{errors are correleated}$$

Steps:

1) Obtain the residuals from the regression model
2) Compute the Durbin-Watson statistic

$$DW = \frac{\sum_{i=2}^{n}(e_i - e_{i-1})^2}{\sum_{i=1}^{n} e_i^2}$$

3) Compare DW to critical value or obtain p-value from software
4) Draw a conclusion.

Note: Rejecting H_0 means the independence assumption is violated.

How to choose the model?

- If including x^2 in the model improves it, why not include x^3, x^4, or even x^5?
- How do we choose the number of knots in spline regression?

What if we choose a very large number to ”be safe”.

- The resulting fit will be unnecessarily wiggly. Why?
- We will run into the overfitting trap.
- How do we know when to stop?

Linear Model Selection

- Ordinary least squares (OLS) regression works well in many real-world applications. Recall, in OLS, we fit a linear model of the form:

$$Y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \ \ldots + \beta_p x_{pi} + \varepsilon_i$$

- We find estimates for the betas that minimize the sum of squared errors (SSE):

$$SSE = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \left(Y_i - \hat{Y}_i\right)^2 = \sum_{i=1}^{n} \left(Y_i - \hat{\beta}_0 - \sum\nolimits_{k=1}^{p} \hat{\beta}_k x_{ki}\right)^2$$

- The solution to the problem can be expressed as:

$$\widehat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{Y}$$

- where the design matrix ***X*** is of size *n* by *p*+1 because the first column is all ones for the intercept.

Linear Model Selection

- In some cases, though, there may be good reasons to use other fitting rules than the normal equations. Why?

  - If the sample size *n* is small and there are a lot of predictors the least squares fit will have high variability.
  - If there are a lot of predictors (i.e., *p* is large), some of them may be *non-informative*. That is, they may not be related to the outcome variable. In that case, OLS will estimate small coefficients for the variables but will not set the coefficients to be exactly zero.
  - Thus, we might seek out a method that sets the magnitude of some coefficients to exactly zero. This would result in *feature selection* or *variable selection*.

- Model specification is the process of determining which independent variables to include and exclude from a regression equation.

Alternatives to OLS for Linear Models

1. **Subset selection.** Instead of fitting OLS on all possible predictor variables, we first eliminate some by setting their coefficients to zero and then fit OLS on the remaining predictors.
2. **Regularization/shrinkage.** The model is fit on all *p* predictors, but the magnitudes of the coefficients are shrunk towards zero. These methods work by adding a penalty to the loss function based on the magnitude of the regression coefficients. One method of regularization, called *the lasso*, uses SSE plus a regularization term as the loss function:

$$\sum_{i=1}^{n}\left(Y_i - \hat{\beta}_0 - \sum_{k=1}^{p}\hat{\beta}_k x_{ki}\right)^2 + \lambda\sum_{j=1}^{p}\left|\hat{\beta}_j\right|$$

SSE: usual error term for OLS regression

Shrinkage penalty for the lasso.

3. Dimension reduction. Here we project the *p* predictors down into an *M*-dimensional space, where *M* < *p*, via linear combinations of variables, or *directions*. We then use the smaller set of *m* projections as the predictors to be fit by OLS.

4. Partial Least Squares (PLS). A *supervised* alternative to PCR.

Subset Selection

- *Best subset selection* involves fitting the OLS model for *every possible combination* of the *p* predictors and picking the one that is best.
- This is a *brute force* method where we must fit all possible models. Assume *p* predictors. The number of possible combinations (order of the predictors doesn't matter) are
- 0 variables $\binom{p}{0} = \frac{p}{0!(p)!}$
- 1 variable: $\binom{p}{1} = \frac{p}{1!(p-1)!}$
- 2 variables: $\binom{p}{2} = \frac{p}{2!(p-2)!}$
- 3 variables: $\binom{p}{3} = \frac{p}{3!(p-3)!}$

  ⋮

  ⋮
- *p* – 1 variables: $\binom{p}{p-1} = \frac{p}{1!(p-1)!}$
- *p* variables: $\binom{p}{p} = \frac{p}{0!(p)!}$

The sum of all these combinations is 2^p.

Best Subset Selection

- Algorithm for best subset selection:
  1. Let M_0 represent the *null model* (i.e., the model with no predictors). This model will only estimate a constant intercept which will represent the sample average.
  2. For $k = 1, 2, \ldots, p$:
     a) Fit all p choose k models that contain exactly k predictors.
     b) Select the best-fitting model among all the p choose k models, as measured by SSE, R^2, or some other measure of model fit, and call it M_k.
  3. Pick the single best model from $M_0, \ldots, M_p$ using cross-validation prediction error, or some other measure of model fit such as the AIC or BIC.

Model Selection Criteria – All Possible Regressions

Suppose we have *k* predictors. Then there are 2^k potential models to compare.
Criteria:

- $R^2 \Leftrightarrow \mathrm{SSE}$: find the model so that R^2 is max, or SSE is min

$$R^2 = 1 - \frac{\mathrm{SSE}}{SST}$$

- $R^2_{\mathrm{adj}} \Leftrightarrow \mathrm{MSE}$: find the model so that R^2_{adj} is max, or MSE is min

$$R^2_{\mathrm{adj}} = 1 - \frac{\mathrm{MSE}}{SST/(n-1)}$$

- Mallow's C_p: find the model with smallest p so that $C_p \leq p$

$$C_p = \frac{SSE_p}{MSE} - n + 2(p+1)$$

- AIC and BIC (discussed later)

Model Validation

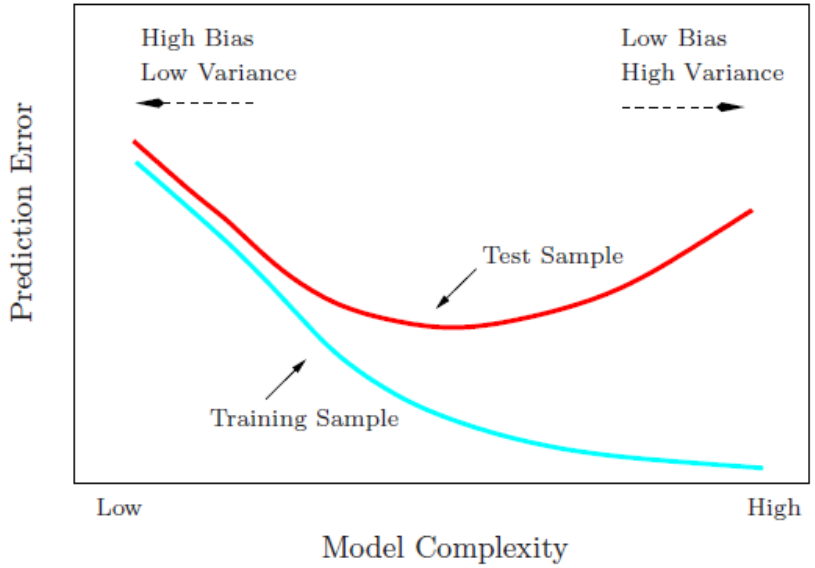
- In statistics, regression validation is the process of deciding whether the numerical results quantifying the relationship between variables are acceptable as description of the data.
- When we have a lot of data, we would like to see how well a model obtained from one set of data (training sample) compares to the predictions on a new set of data (validation or test sample), and how the training model fits the new data.
- Models should give similar fits based on SSE_p, $PRESS_p$, C_p, MSE_p and regression coefficients
- Mean Square Error (MSE) when training model is applied to validation sample of size n^*:

$$MSE = \frac{\sum_{i=1}^{n^*}\left(Y_i - \hat{Y}_i\right)^2}{n^*}$$

where $\hat{Y}_i = b_0^T + b_1^T X_{i1}^V + \cdots + b_p^T X_{i,p}^V$

MSE

- A good learner is the one which has good prediction accuracy; in other words, which has the smallest prediction error.
- The MSE measured on the *training* set is an underestimate of the test set MSE.
- In particular, the training set MSE will go down so long as more variables/flexibility are included in the model; however, the *test* set MSE will not.



Probabilistic selection: AIC and BIC

- The Akaike Information Criterion, or AIC for short, is a method for scoring and selecting a model. It is named for the developer of the method, Hirotugu Akaike, and may be shown to have a basis in information theory and frequentist-based inference. BIC is a similar concept.
- The AIC and BIC are both based on -2 times the maximized value of the loglikelihood (-2*LL*); *k* is the number of parameters and *LL* is the log-likelihood value at the MLE:

$$\text{AIC} = 2k/n - 2LL/n, \ \text{BIC} = \ln(n)*k - 2LL$$

Here there is a penalty for magnitude of the *LL*; smaller is better

Here there is a penalty for more complexity in the model.

- The likelihood (assuming normally distributed errors) for multiple linear regression is

$$\left(\frac{1}{\sqrt{2\pi}\sigma}\right)^n e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}\left(Y_i - \boldsymbol{\beta}'\boldsymbol{x_i}\right)^2}$$

- The score, as defined above, is minimized, e.g. the model with the lowest AIC is selected.

AIC and BIC

- As you see from their definitions, AIC and BIC are similarly constructed, with the essential difference being the term multiplied by *k*, the number of estimated coefficients.
- As a result (2 vs. log(n)), the BIC tends to penalize model complexity more heavily than the AIC.
- The AIC tries to select the model that most adequately describes an unknown, high dimensional reality.
- Because the AIC is somewhat more permissive of model complexity than the BIC, it may be preferred for *prediction*.
- When the intent is to create a model for *explanation*, BIC may be preferred because it will tend to produce a simpler (i.e., more easily interpretable) model.
- Importantly, the derivation of BIC under the Bayesian probability framework means that if a selection of candidate models includes a true model for the dataset, then the probability that BIC will select the true model increases with the size of the training dataset. This cannot be said for the AIC score.

Best Subset and Computational Efficiency

- Best subset selection is very *computationally demanding*.
- For p = 30, for example, best subset will require fitting 2^{30} = 1,073,741,824 models.
- Whereas forward selection (described next) will require fitting 1 + 1 + 2 + 3 + … + 29 + 30 = $p(p + 1)/2 + 1$ = 451 models.
- That said, the ease of fitting Gaussian linear models make even large (i.e., 30+ covariates) problems tractable within a few seconds. This is possible in part due to advances in computational algorithms over the last decades.
- The problem of computational efficiency is quickly made worse, however, by using other models that require iterative numerical methods for solving.
- For example, with Gaussian linear models (which can be solved analytically), a best subset search with 11 predictors takes less than a second.
- By contrast, with a logistic model (solved using Newton's method), 11 predictors takes nearly 2 minutes, and adding each subsequent predictor causes the time to more than double.

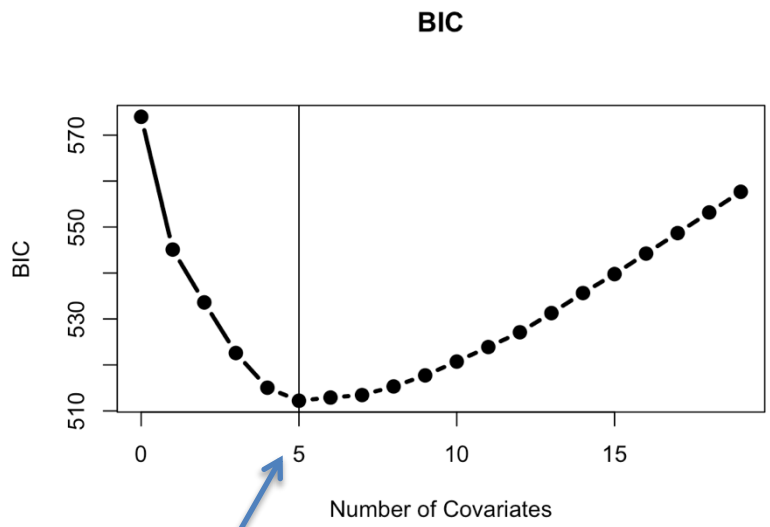
Example

Ninety members (ages 18.1 to 23.4 years) of three Division I women's intercollegiate rowing teams (National Collegiate Athletic Association) within the Big Ten Conference volunteered to participate in a study to predict race time for female collegiate rowers from nineteen physical characteristics.

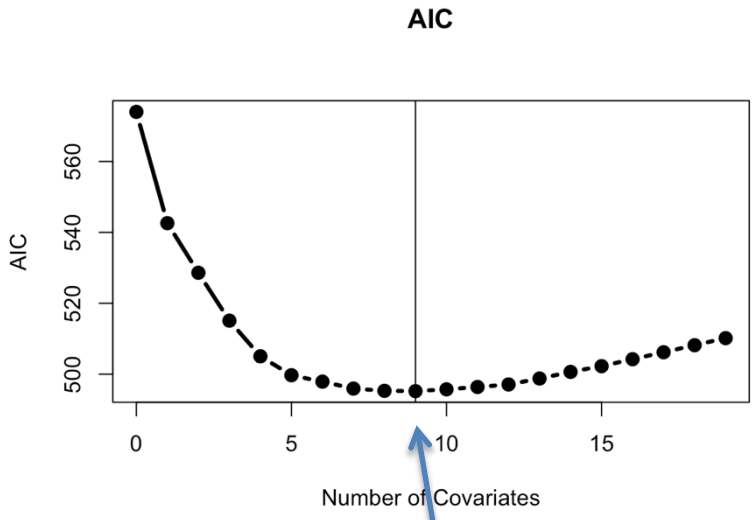
```
racetime          tall           weight          armspan         flexarm         thighci         calfcir
Min.   :406.0   Min.   :155.6   Min.   : 57.00   Min.   :138.5   Min.   :27.50   Min.   :50.50   Min.   :34.0
1st Qu.:442.4   1st Qu.:169.3   1st Qu.: 69.90   1st Qu.:167.6   1st Qu.:30.50   1st Qu.:55.50   1st Qu.:37.0
Median :460.0   Median :173.6   Median : 73.35   Median :172.7   Median :31.90   Median :57.50   Median :38.0
Mean   :458.5   Mean   :172.9   Mean   : 74.62   Mean   :172.8   Mean   :31.60   Mean   :57.89   Mean   :38.4
3rd Qu.:475.3   3rd Qu.:176.3   3rd Qu.: 79.10   3rd Qu.:177.5   3rd Qu.:32.88   3rd Qu.:59.88   3rd Qu.:39.5
Max.   :518.7   Max.   :186.4   Max.   :100.10   Max.   :205.5   Max.   :39.00   Max.   :71.50   Max.   :44.0
    tricep          biceps          thigh           estffm          estfm           bestsnr         bestvj
Min.   :12.00   Min.   : 4.00   Min.   :13.00   Min.   :44.86   Min.   : 9.918   Min.   :17.00   Min.   : 8.00
1st Qu.:18.00   1st Qu.: 9.00   1st Qu.:24.00   1st Qu.:54.52   1st Qu.:14.274   1st Qu.:34.00   1st Qu.:13.00
Median :21.00   Median :12.00   Median :28.00   Median :56.97   Median :16.929   Median :40.00   Median :15.00
Mean   :21.09   Mean   :12.46   Mean   :28.73   Mean   :57.83   Mean   :16.786   Mean   :38.72   Mean   :14.82
3rd Qu.:24.00   3rd Qu.:15.00   3rd Qu.:34.00   3rd Qu.:61.15   3rd Qu.:19.073   3rd Qu.:44.00   3rd Qu.:16.00
Max.   :34.00   Max.   :28.00   Max.   :50.00   Max.   :78.15   Max.   :26.726   Max.   :50.00   Max.   :21.00
   legpower          endo            meso            ecto          expvarsity        preexper
Min.   : 60.19   Min.   :3.219   Min.   :1.564   Min.   :0.100   Min.   :0.0000   Min.   :0.0000
1st Qu.: 92.19   1st Qu.:4.768   1st Qu.:3.257   1st Qu.:1.004   1st Qu.:0.0000   1st Qu.:0.0000
Median : 99.84   Median :5.789   Median :4.033   Median :1.596   Median :1.0000   Median :0.0000
Mean   :100.33   Mean   :5.696   Mean   :4.042   Mean   :1.655   Mean   :0.5667   Mean   :0.3333
3rd Qu.:108.92   3rd Qu.:6.499   3rd Qu.:4.738   3rd Qu.:2.218   3rd Qu.:1.0000   3rd Qu.:1.0000
Max.   :153.98   Max.   :8.260   Max.   :6.812   Max.   :3.887   Max.   :1.0000   Max.   :1.0000
```

Rowing Example – Best Subset Selection

```
# Outcome is racetime (in seconds)
# 19 predictors examined
bs3 <- bestglm(Xy = d, family = gaussian, IC = "BIC")
bs4 <- bestglm(Xy = d, family = gaussian, IC = "AIC")
```



BIC identified 5 covariates as the optimal number.



AIC identified 9 covariates as the optimal number.

Important: Best subset selection works here because the model is Gaussian. If we were to try a logistic model for binomial data, the max number of predictors is 15.

Row time Example - Results

BIC Results

	Estimate	Std. Error	t value	Pr(>|t|)	
(Intercept)	979.6078	60.0014	16.326	< 2e-16	***
tall	-3.3133	0.3837	-8.636	3.20e-13	***
calfcir	3.0876	1.1565	2.670	0.00911	**
meso	-13.1093	2.6979	-4.859	5.41e-06	***
expvarsity	-17.8388	3.3819	-5.275	1.02e-06	***
preexper	-11.1860	3.6139	-3.095	0.00267	**

AIC Results

	Estimate	Std. Error	t value	Pr(>|t|)	
(Intercept)	797.8807	108.6409	7.344	1.54e-10	***
tall	-2.2842	0.6347	-3.599	0.000553	***
calfcir	2.7636	1.1699	2.362	0.020592	*
biceps	1.1183	0.5703	1.961	0.053377	.
estfm	1.6653	1.2028	1.384	0.170056	
bestvj	5.1014	1.7652	2.890	0.004957	**
legpower	-1.1402	0.4546	-2.508	0.014161	*
meso	-9.7856	3.3879	-2.888	0.004979	**
expvarsity	-18.3696	3.4565	-5.315	9.42e-07	***
preexper	-11.0965	3.5822	-3.098	0.002691	**

Notice that overall, the effects have the same signs, but their magnitudes are different.

Regression Model Building & Diagnostics

- Outliers
- Influential points
- Variance inflation factor
- Shrinkage and regularization

Outlying *Y* Observations – Studentized Residuals

- Recall the model errors (unobservable):

$$\varepsilon_i = Y_i - (\beta_0 + \beta_1 X_{i1} + \ldots + \beta_p X_{ip})$$

such that $E(\varepsilon_i) = 0$, $Var(\varepsilon_i) = \sigma^2$, $cov(\varepsilon_i , \varepsilon_j) = 0$

- We estimate the errors with the (raw) observed residuals (in R: `resid()`):

$$e_i = Y_i - \hat{Y}_i = Y_i - \left(b_0 + b_1 X_{i1} + \ \ldots + b_p X_{ip}\right)$$

such that $E(e_i) = 0$, $Var(e_i) = (1 - h_{ii})\sigma^2$, $cov(e_i , e_j) = -h_{ij}\, \sigma^2$
where h_{ij} is the $(i, j)^{th}$ element of $\boldsymbol{H} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$
and we estimate $s^2(e_i) = (1 - h_{ii})MSE$, $s(e_i, e_j) = -h_{ij}\, MSE$

- Semi-studentized residual is

$$e_i^* = \frac{e_i}{\sqrt{MSE}}$$

- Studentized residual (in R: `rstandard()`):

$$r_i = \frac{e_i}{\sqrt{MSE(1 - h_{ii})}}$$

Outlying *Y* Observations – Studentized Deleted Residuals

Deleted Residual (Observed value minus fitted value when regression is fit on the other n-1 cases):

$$d_i = Y_i - \hat{Y}_{i(i)} \qquad \hat{Y}_{i(i)} = b_{0(i)} + b_{1(i)}X_{i1} + \ldots + b_{p-1(i)}X_{i,p-1}$$

$b_{k(i)} \equiv$ regression coefficient of X_k when case i is deleted

Studentized Deleted Residual (makes use of having predicted i^{th} response from regression based on other n-1 cases):

$$t_i = \frac{d_i}{s\{d_i\}} \sim t(n-p-1)$$

$$s^2\{d_i\} = s^2\{\text{pred}_i\} = MSE_{(i)}\left[1 + \mathbf{X_i}'\left(\mathbf{X_{(i)}}'\mathbf{X_{(i)}}\right)^{-1}\mathbf{X_i}\right] \qquad \mathbf{X_i}' = \begin{bmatrix} 1 & X_{i1} & \text{L} & X_{i,p-1} \end{bmatrix}$$

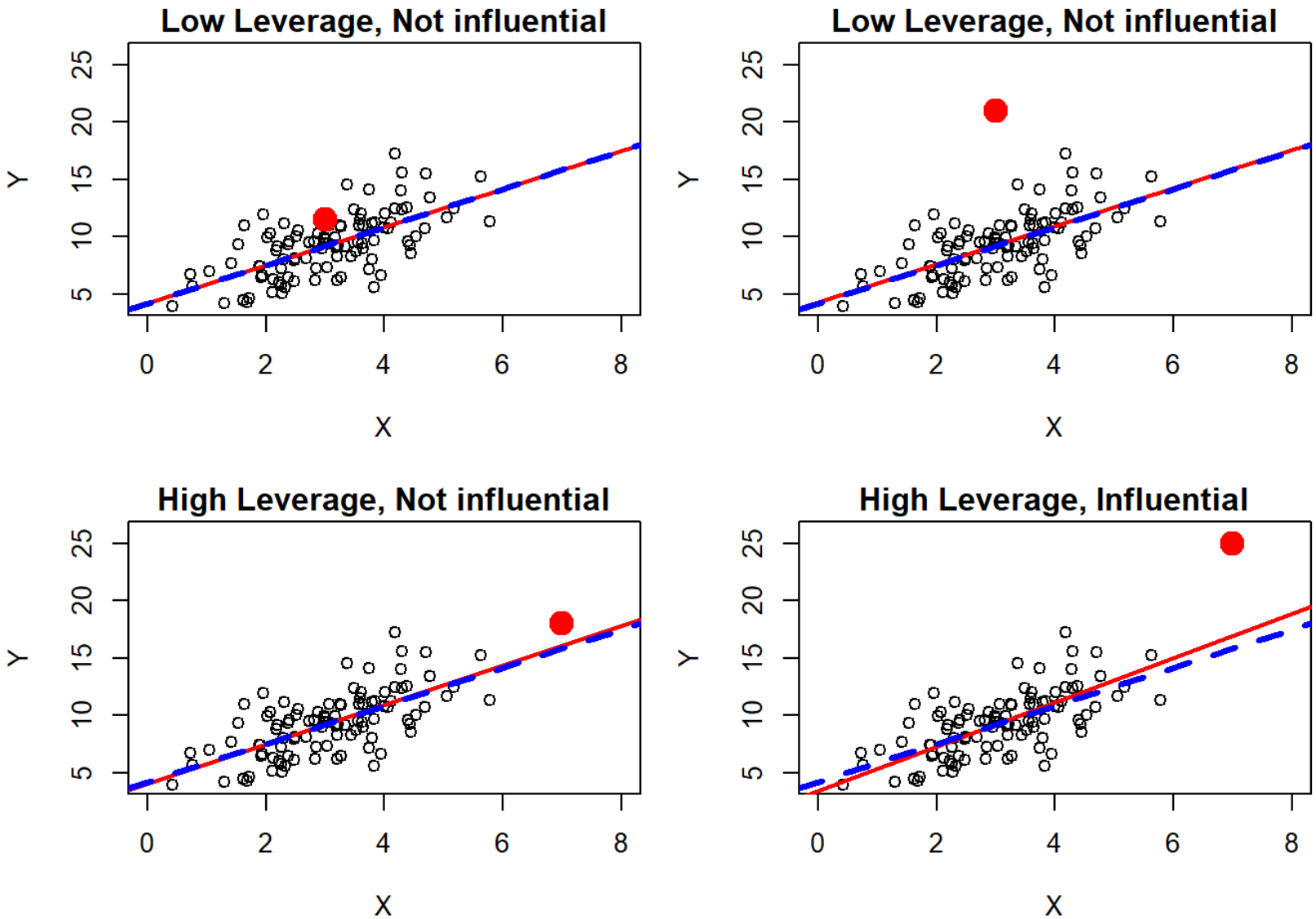
Note: $SSE = (n-p)MSE = (n-p-1)MSE_{(i)} + \dfrac{e_i^2}{1-h_{ii}}$

$$\Rightarrow \quad t_i = e_i\left[\frac{n-p-1}{SSE(1-h_{ii}) - e_i^2}\right]^{1/2}$$

Computed without re-fitting n regressions

Test for outliers (Bonferroni adjustment): Outlier if $|t_i| \geq t\left(1 - \left(\dfrac{\alpha}{2n}\right), n-p-1\right)$

Leverage vs Influential Points



Outlying *X*-Cases – Hat Matrix Leverage Values

Recall the *n* x *n* "hat" matrix

$$\boldsymbol{H} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$$

Properties:

The diagonal elements are $0 \leq h_{ii} \leq 1$

$$tr(\boldsymbol{H}) = \sum_{i=1}^{n} h_{ii} = tr(\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}') = tr\left(\boldsymbol{X}'\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-\mathbf{1}}\right)$$

$$= tr\left(\boldsymbol{I}_{p+1}\right) = p + 1$$

Cases with *X*-levels close to the "center" of the sampled *X*-levels will have small leverages.

Cases with "extreme" levels have large leverages and have the potential to "pull" the regression equation toward their observed Y-values.

Rule of thumb: Large leverage values are those > 2(p+1)/n (or more than two times larger than the average)

Identifying Influential Cases – Fitted Values

Influential cases in terms of their own fitted values:

$$DFFITS_i = \frac{\hat{Y}_i - \hat{Y}_{i(i)}}{\sqrt{MSE_{(i)} h_{ii}}}$$

Represents the # of standard errors own fitted value is shifted when case included vs. excluded

Shortcut computational formula:

$$DFFITS_i = e_i \left[\frac{n-p-2}{SSE(1-h_{ii}) - e_i^2}\right]^{\frac{1}{2}} \left(\frac{h_{ii}}{1-h_{ii}}\right)^{\frac{1}{2}} = t_i \left(\frac{h_{ii}}{1-h_{ii}}\right)^{\frac{1}{2}}$$

Problem cases when:

$$\boxed{(DFFITS)_i > 2\sqrt{\frac{p+1}{n}}}$$

Simpler rule:

$(DFFITS)_i > 1$ for datasets with $n < 30$.

Identifying Influential Cases – Fitted Values

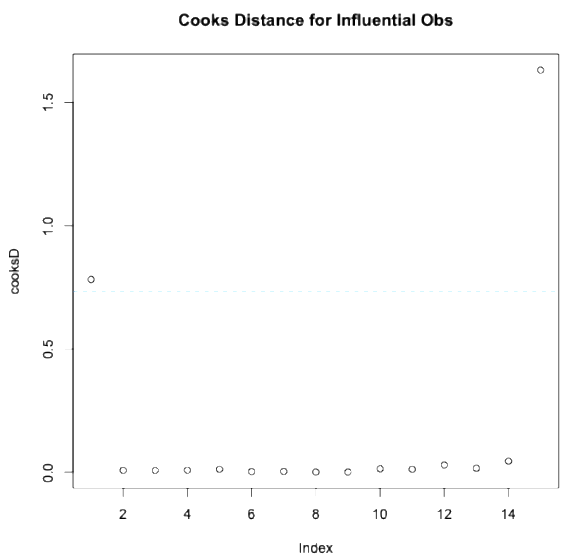
Influential cases in terms of all fitted values – Cook's Distance:

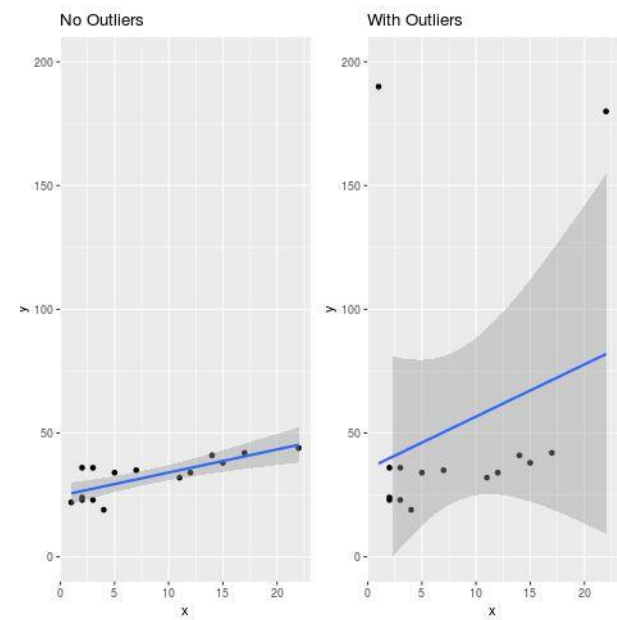
$$D_i = \frac{\sum_{j=1}^{n}\left(\hat{Y}_j - \hat{Y}_{j(i)}\right)^2}{(p+1)MSE} = \frac{e_i^2}{(p+1)MSE}\frac{h_{ii}}{(1-h_{ii})^2}$$

Problem cases when:

$$\boxed{D_i > F(0.5, p+1, n-p-1)}$$

Simpler rule for large n: $D_i > 1$ or $D_i > 4/n$ for small n.





Influence on the Regression Coefficients

A measure of the influence of the ith case on each regression coefficient b_k is the difference between the estimated regression coefficient b_k based on all n cases and the regression coefficient $b_{k(i)}$ obtained when the ith case is omitted:

$$(\text{DFBETAS})_{k(i)} = \frac{b_k - b_{k(i)}}{\sqrt{MSE_{(i)} c_{kk}}}$$

where

$$(\boldsymbol{X}'\boldsymbol{X})^{-1} = \begin{bmatrix} c_{00} & c_{01} & \cdots & c_{0p} \\ c_{10} & c_{11} & \cdots & c_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ c_{p0} & c_{p1} & \cdots & c_{pp} \end{bmatrix}$$

Problem cases when:

$$(DFBETAS)_{k(i)} > \frac{2}{\sqrt{n}}$$

Note: Cook's Distance can also be computed as aggregate measure on the the influence on all regression coefficients and is algebraically equivalent to the overall influence on all fitted values.

Multicollinearity - Variance Inflation Factors

- Problems when predictor variables are correlated among themselves
  - Regression Coefficients of predictors change, depending on what other predictors are included
  - Extra Sums of Squares of predictors change, depending on what other predictors are included
  - Standard Errors of Regression Coefficients increase when predictors are highly correlated
  - Individual Regression Coefficients are not significant, although the overall model is
  - Width of Confidence Intervals for Regression Coefficients increases when predictors are highly correlated
  - Point Estimates of Regression Coefficients are wrong sign (+/-)

Multicollinearity – Diagnostics

- Large changes in the estimated regression coefficients when a predictor variable is added or deleted.
- Nonsignificant results in t-tests on the regression coefficients.
- Estimated regression coefficients with an algebraic sign that is the opposite of that expected.
- Large coefficients of simple correlation between pairs of predictor variables in the correlation matrix.
- Strong linear associations between pairs of predictors in the scatterplot matrix.
- Formal diagnostics with Variance Inflation Factor (VIF) for var. X_k:

$$VIF_k = \frac{1}{1 - R_k^2}$$

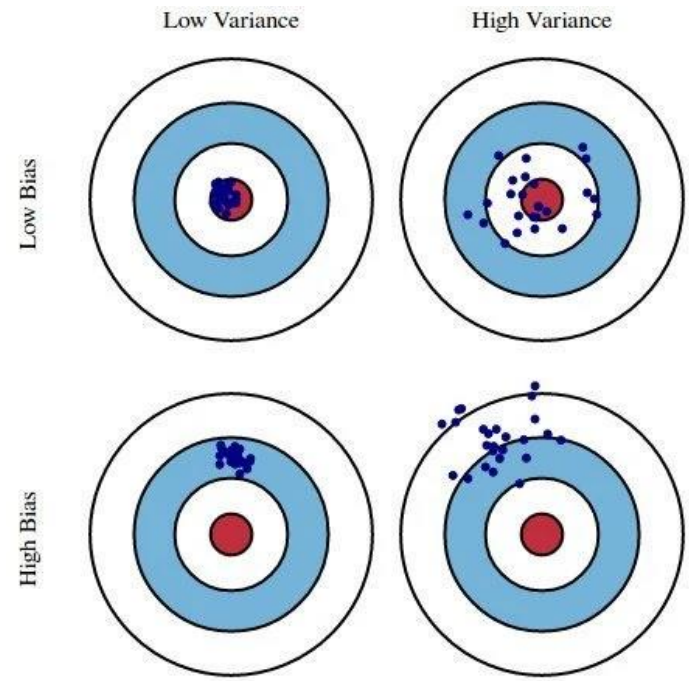
where R_k^2 is from regressing X_k on the other predictors.

- If $VIF_k > 10$ or $\overline{VIF} > 5$, then we have a multicollinearity issue.

Multicollinearity Remedy – Shrinkage Methods

Both the bias and the variance are desired to be low, as large values result in poor predictions from the model. In fact, the model's error can be decomposed into three parts: error resulting from a large variance, error resulting from significant bias, and the remainder - the unexplainable part.

$$E(e) = (E(X\hat{\beta}) - X\beta)^2 + E(X\hat{\beta} - E(X\hat{\beta}))^2 + \sigma^2 = Bias^2 + Variance + \sigma^2$$



Ridge Regression

- Ridge regression is a way to create a parsimonious model when the number of predictor variables exceeds the number of observations, or when a data set has multicollinearity.
- The essential idea with ridge regression is identical to OLS regression with one exception, there is a penalty added in the form of the l_2 norm of the coefficients.
- The l_2 *norm* is defined as follows. For $\mathbf{x} = (x_1, x_2, \ldots, x_n)$, the l_2 norm of $\mathbf{x}$ (also called the Euclidean norm) is

$$\|\boldsymbol{x}\|_2 = \sqrt{\sum_{i=1}^{n} x_i^2}$$

- The ridge regression coefficients $\hat{\beta}_{j,\lambda}^{R}$ are the betas that minimize

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}\beta_j^2 = SSE + \lambda\|\boldsymbol{\beta}\|_2^2$$

where $\lambda \geq 0$ is a *tuning parameter*.